RUSHIL GUPTA

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EDUCATION

Ashoka University, *B.Sc (Hons) in Mathematics and Computer Science with Research* Expected 2027

- CGPA: 3.91/4, Major CGPA: 4.0/4.0.
- Relevant Coursework: Reinforcement Learning, Machine Learning, Fourier Series, Linear Algebra and Matrix Analysis, Measure Theory, Advanced Algorithms, Cryptography, Stochastic Calculus and Mathematical Finance.
- Teaching Assistantships: Reinforcement Learning, Introduction to Machine Learning, Numerical Algorithms and Optimisation, Introduction to Computer Science.

PUBLICATIONS

A Splitting Method for SDE Terminal-Law Estimation, *MLxOR @ NeurIPS 2026* Sep 2026

- Path-splitting method for SDE sampling under a fixed budget; 12–26% lower KS error than i.i.d. sampling.

Fundamental limits for weighted empirical approximations of tilted distributions, *AISTATS 2026* Jan 2026

- Provided scaling limits for empirical reweighted estimators, characterizing regimes where the estimator is accurate.

RESEARCH EXPERIENCE

Stanford University, *Undergraduate Visiting Intern* Jul 2026 – Aug 2026

- Working with Prof. Peter Glynn on using ideas from Empirical Bayes to improve estimates in simulation models.

Safexpress Centre for Data, Learning and Decision Sciences, *Undergraduate Researcher* Aug 2024 – Present

- Developed a provably efficient estimator for simulating stochastic differential equations under a fixed budget.
- Analyzed the sample complexity of diffusion models for accurately modeling reweighted distributions.
- Developed methods that use small agent-based models to approximate the spread of disease in larger models.

Joint Cipher Bureau (DRDO), *Research Intern* Jan 2026 – Present

- Studied the mathematical foundations of post-quantum cryptography, from lattice problems to lattice-based IBE.
- Built a scheme-agnostic PKI for offline post-quantum key distribution, with a KDC issuing lab certificates for mTLS.

KOITA Centre for Digital Health Ashoka, *Research Intern* Feb 2024 – Dec 2024

- Developed infrastructure for a cohort study integrating data streams from multiple wearables and a custom app.
- Worked on an LLM solution to automate informed consent, leveraging agentic models for improved performance.

WORK EXPERIENCE

ML Researcher, *Truth Audit Labs* May 2026 – Jun 2026

- Generated synthetic data and trained a policy monitor to flag policy-violating text in organizational messages.
- Red-teamed LLM assistants via prompt injection to override or leak system prompts and elicit harmful outputs.

Platform Teams Intern, *Sarvam AI* Nov 2024 – Jan 2025

- Designed and implemented a versioning system for voice agents for incremental updates and seamless deployment.
- Part of the core team that owned the agents platform frontend and maintained several of its backend microservices.
- Implemented the first unit and integration tests for backend microservices, improving API reliability.

POSITIONS OF RESPONSIBILITY

Head of Events, *Ashoka Computer Science Society* Sep 2024 – May 2025

- Led a team of 20 members in planning and executing events, including talks, workshops, and mixers.
- Initiated Leetcode Saturdays; hosted weekly competitive programming sessions for students to solve problems.

Co-Founder and Head of Events & Logistics, *Ashoka Health Research Club* Mar 2024 – Dec 2024

- Ideated and co-founded the club, being involved in the selection process of the core and events team.
- Established a framework connecting students with faculty research, offering observership and internship tracks.

SERVICE AND ACTIVITIES

Reviewer, *ICLR 2027* 2026

Squash Team Member, *Ashoka University* 2023 – Present